2.3 A Divergent Series and Its Convergent Cousin

WORKSHEET

The question

STEP III 2014, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q8

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The first inequality is about the block of terms from $r = k^n$ to $r = k^{n+1} - 1$. With $k = 2$ and $n = 3$, which terms are these, and how many of them are there?

STAGE 1 Bounding a block

Covers: the first inequality

Ask yourself With $k = 2$ and $n = 3$, which terms make up the block from $r = k^n$ to $r = k^{n+1} - 1$, and how many are there? In general, how many terms are in the block?

Pause and try f is decreasing. What are the smallest and largest values of $f(r)$ in the block? Multiply each by the number of terms.

STAGE 2 The harmonic series

Covers: Part (i)

Ask yourself With $k = 2$ and $f(r) = \frac{1}{r}$, work out the two bounds for the block from 2^n to $2^{n+1} - 1$.

Pause and try Every block contributes at least the same fixed amount. Add the blocks for $n = 0, 1, \ldots, N$: what happens to the partial sums as $N \to \infty$?

STAGE 3 The sum of reciprocal cubes

Covers: Part (ii)

Ask yourself Now take $f(r) = \frac{1}{r^3}$, still with $k = 2$. What is the upper bound for block n?

Pause and try The blocks for $n = 0, 1, 2, \ldots$ cover every positive integer once. Add up the upper bounds: it is a geometric series.

STAGE 4 Counting numbers without a 2

Covers: Part (iii)

Pause and try Write every number from 0 to 999 with three digits, allowing leading zeros (so 7 is 007). How many choices are there for each digit if 2 is not allowed? Remember to remove 0.

STAGE 5 Kempner's series

Covers: Part (iii)

Ask yourself Use blocks defined by the number of digits; this is like taking $k = 10$.

Pause and try How many d-digit numbers have no 2 in them? (The first digit cannot be 0.) What is the smallest a d-digit number can be? Bound the contribution of each block and add over d.

What's really going on

The condensation test

The first inequality in the question is the heart of *Cauchy's condensation test*: if f is positive and decreasing, then

$$\sum_{r=1}^{\infty} f(r) \text{ converges } \iff \sum_{n=0}^{\infty} 2^n f(2^n) \text{ converges.}$$

It turns hard series into easy ones. Applied to $f(r) = 1/r^p$, it gives $\sum 2^n \cdot 2^{-np} = \sum (2^{1-p})^n$, a geometric series that converges exactly when $p > 1$. Applied to $1/(r \ln r)$ it shows that even this diverges, and to $1/(r(\ln r)^2)$ that this converges. The argument goes back to Nicole Oresme, who proved the harmonic series diverges in about 1350 by exactly the blocks of Figure 2.3.1 (in the full section online).

How slowly the harmonic series diverges

Figure 2.3.1 (in the full section online) suggests that each block contributes about $\ln 2$, so the sum of the first N terms should be roughly $\log_2 N \times \ln 2 = \ln N$. In fact

$$1 + \frac{1}{2} + \cdots + \frac{1}{N} = \ln N + \gamma + (\text{something tending to } 0),$$

where $\gamma = 0.5772\ldots$ is the Euler–Mascheroni constant. Nobody knows whether γ is rational. You will prove the formula with integrals in first-year analysis, and the harmonic numbers turn up again in Section 9.11 and in Section 11.12.

Most numbers contain a 2

Why does Kempner's series converge? Because "numbers without a 2" are very rare among large numbers: of the numbers below 10^d, the proportion with no 2 is at most $(0.9)^d$, which tends to 0. Among 100-digit numbers, fewer than 0.003% avoid the digit 2. So we are not removing a tenth of the terms, but almost all of them. Our bound of 80 is generous: in 1979 Robert Baillie used an accelerated method to compute the sum to many decimal places, and it is about 19.257.

Sums of reciprocal powers

The sum in part (ii) is $\zeta(3)$, a value of the Riemann zeta function we met in Section 1.2. In Section 2.4 we will find $\zeta(2) = \pi^2/6$ exactly, and Euler found a formula for $\zeta(4), \zeta(6), \ldots$. But nobody has found a formula for $\zeta(3)$. In 1978 Roger Apéry stunned the mathematical world by proving that $\zeta(3)$ is irrational, using integrals not unlike the ones in Section 2.9. Whether $\zeta(5)$ is irrational is still unknown.

Try this next

1. **Feel the slowness.** Using $\sum_{r \le N} 1/r \approx \ln N + 0.5772$, estimate how many terms of the harmonic series you need to pass 10, and to pass 100.

2. **The test in action.** Use the method of part (i) with $f(r) = 1/r^2$ to show that $\sum 1/r^2 \le 2$. Then improve the bound by treating the first term separately.

3. **Between convergence and divergence.** Use the block inequality with $k = 2$ to show that $\sum_{r \ge 2} \dfrac{1}{r \ln r}$ diverges and $\sum_{r \ge 2} \dfrac{1}{r (\ln r)^2}$ converges.

4. **Excluding more.** HARDERShow that the sum of the reciprocals of the positive integers with no 2 *and* no 7 in their decimal representation is less than 35. (Can you see where 35 comes from?)

Further reading

► **Julian Havil, *Gamma: Exploring Euler's Constant*** (Princeton University Press). The

harmonic series, γ and much more, written for a strong school student.

- **Mary Hart, *Guide to Analysis*** (Palgrave). A concise first-year text with a clear chapter on series and comparison tests.
- **Robert Baillie, "Sums of reciprocals of integers missing a given digit"**, *American Mathematical Monthly* 86 (1979). A short paper you can read with this section behind you.

The full guided solution is in Section 2.3:
https://beyondstep.keithmcnulty.org/ch02/kempner.html